



DESIGN REPORT : TEAM GARUDA

Team Name: Garuda

Aircraft Name: Garuda 2.0

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1 DESIGN OVERVIEW

1.1 Aircraft configuration type

The aircraft utilizes a high-wing monoplane configuration with a conventional tail. This layout was chosen for its inherent longitudinal stability and self-righting tendencies, which are critical for maintaining a steady flight path during payload delivery missions.

1.2 Overall design concept

Our design philosophy prioritizes a balance between payload capacity and structural rigidity. Using an iterative approach, we selected the **CH-10 airfoil** for its superior lift characteristics at low speeds. The fuselage is designed as a streamlined rectangular box to minimize drag while providing a large internal volume for the 250g cube payload.

1.3 How our design addresses the competition requirements

- **Dimensions:** The 1.18 m wingspan and 0.9m length comply with all AeroVerse size constraints.
- **Payload:** The internal bay is centered exactly at the CG to ensure stability remains unaffected during the drop.
- **Safety:** The high-wing placement and 3-degree dihedral ensure a high static margin for safe operation.

2 TECHNICAL SPECIFICATIONS

- **Wingspan:** 1.18 m
- **Aircraft weight without payload:** ≈ 921 g
- **Propeller diameter:** $11'' \times 6$
- **Motor specifications:** 1500KV BLDC Motor
- **Battery details:** 3S 2200mAh LiPo (11.1V)

2.1 Dimensions

1. **Fuselage:** $L : 0.9m, W : 0.13m, H : 0.13m$
2. **Wing:** Span: 1.18m, Chord: 0.28m, Dihedral: 3°
3. **Tail (Horizontal):** Span: 0.52m, MAC : 0.1305m
4. **Tail (Vertical):** Height: 0.19m, MAC : 0.0963m

2.2 Propulsion System

Component	Specification	Justification
Motor	1500KV BLDC	High torque for cargo lift
ESC	40A Brushless	Reliable current handling for 3S
Battery	3S LiPo (2200 mAh)	Optimal weight-to-energy ratio

3 DESIGN CALCULATIONS

- **Thrust-to-Weight Ratio:** Calculated at 1.1 (with payload) and 0.9 (without payload), ensuring climb performance.
- **Wing Loading:** 4.19 kg/m^2 (approx. 41 N/m^2), allowing for a very low stall speed.
- **Airfoil Selection:** CH-10. Chosen for high C_L and predictable stall behavior in cargo-lift applications.
- **Center of Gravity (CG):** Positioned at 25% – 30% of the wing MAC ($70mm - 84mm$ from LE).
- **Static Margin:** $\approx 10\%$, providing positive longitudinal stability.

4 PAYLOAD SYSTEM DESIGN

4.1 Payload compartment design

The compartment is a balsa cage structure designed to house a 250g cube with push springs attached at the roof of the cage for ease of drop.

4.2 RC-operated mechanism

A high-torque 9g servo operates a trap door mechanism which opens and drops the payload inside it.

5 MATERIALS AND CONSTRUCTION

- **Fuselage:** Depron foam skin.
- **Wing:** Laser-cut Balsa ribs with a Balsa wood main spar for bending stiffness.
- **Assembly:** The wing is mounted via rubber bands to the fuselage ease of operation and crash-repair ease.

6 COMPETITION STRATEGY

- **Qualifier Round:** Focus on maximum flight time and stability with the 250g payload.
- **Main Round:** Prioritizing precision in the drop zone by timing the release based on ground-speed estimates.
- **Safety:** Pre-flight checks include range testing and verification of the fail-safe return-to-home protocol.

7 CONSTRUCTION PHOTOS AND CAD FILE

https://drive.google.com/drive/folders/1Txp1gwdNweSBexzmWJwU87o00N__N0AG?usp=sharing